

Recursion - Permutations

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Permutations

A permutation is an arrangement of all element of a set in every possible order.

Example:- "abc"

permutation:- "abc"; "acb"; "bac"; "bca"; "cab"; "cba"

Idea :- str = abc

processed - ϕ / "abc" $\rightarrow$ unprocessed.

$\downarrow$
a / "bc"

$\downarrow$ $\downarrow$
"ab/c" ba/c"

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
abc/ ϕ acb/ ϕ cab/ ϕ bac/ ϕ bca/ ϕ cba/ ϕ

$\Rightarrow$ Pick a character

$\Rightarrow$ Fix it in all the possible character positions

Base Case :- if the string is empty $\rightarrow$ just return.

Note :- Here no of recursive call is variable.

No of function call = length of unprocessed + 1.